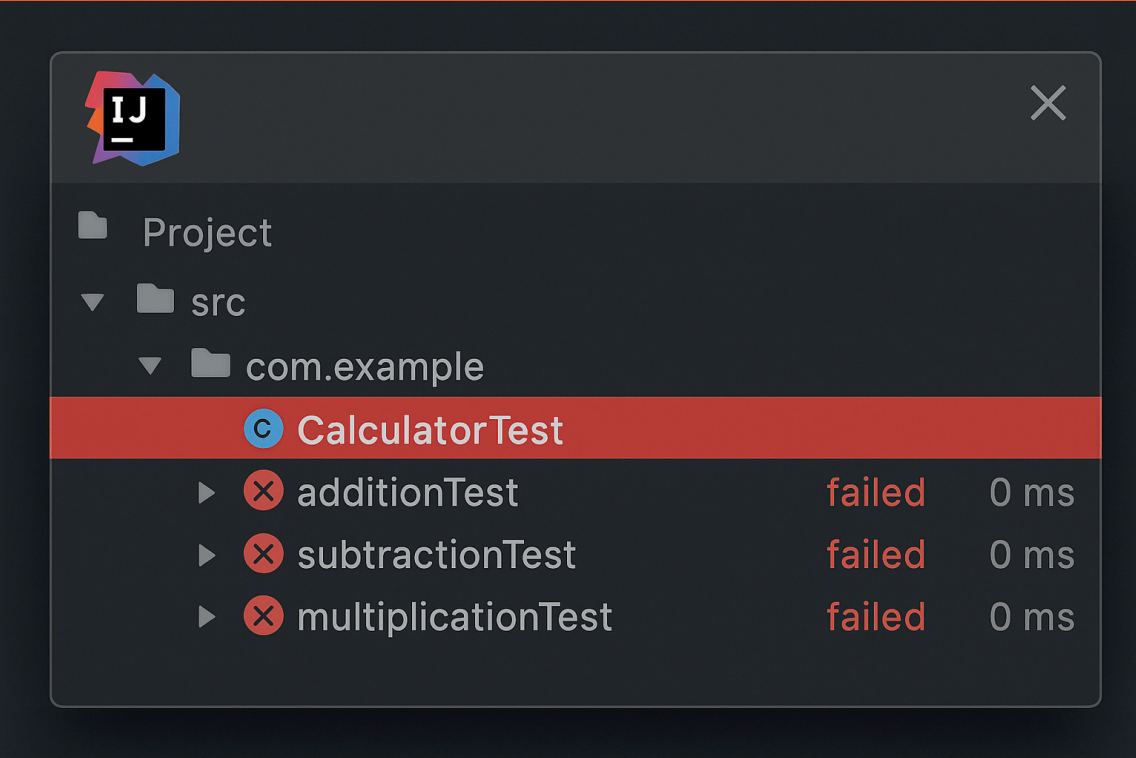


TEST REPAIR AGENT

Research project, focused on repairing broken test cases via a simple Agent in a preset environment. Achieves ~95% success ratio on the dataset of ~450 broken test methods. Evaluated for 3 different models to get the best match accuracy with the original solution.



LOOKS FAMILIAR?



Fixing all the tests after slight code modification...

SOUNDS BORING?

MADE BY
SOFTWARE TESTING
RESEATCH TEAM

TEST REPAIR AGENT

Fixes outdated tests with
95% success ratio



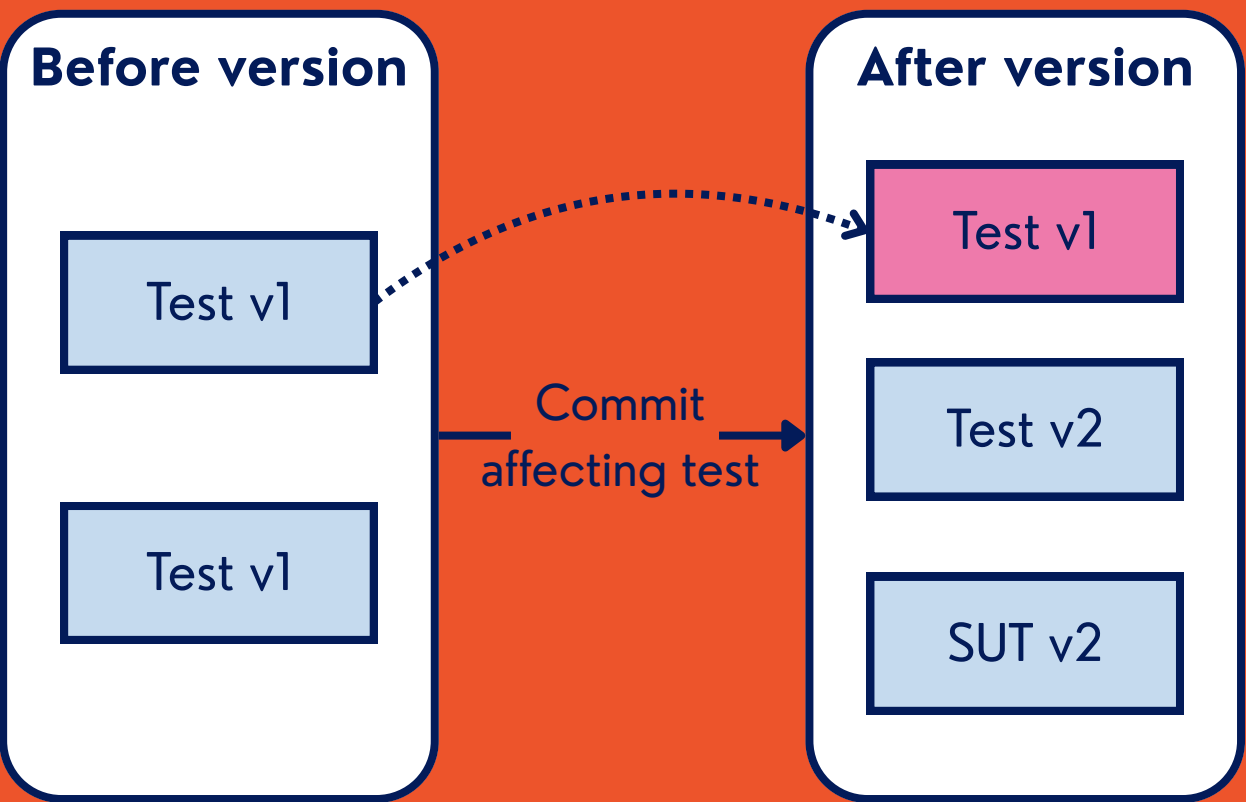
FIRST PROTOTYPE AT AGENT

- Tools:
1. Project exploration
 2. Test modification
 3. Test execution

EVALUATION PIPLINE

DATA COLLECTION	SETUP	AGENT	EVALUATION
Test case	Environment preparation	Test repairing	Result processing
Project ID	Initialize environment	Context collection	Validate repairs
Test ID	Setup project	Test modification	Calculate metrics
LLM Model	Setup broken test	Test execution	Generate reports

DATA COLLECTION & SETUP

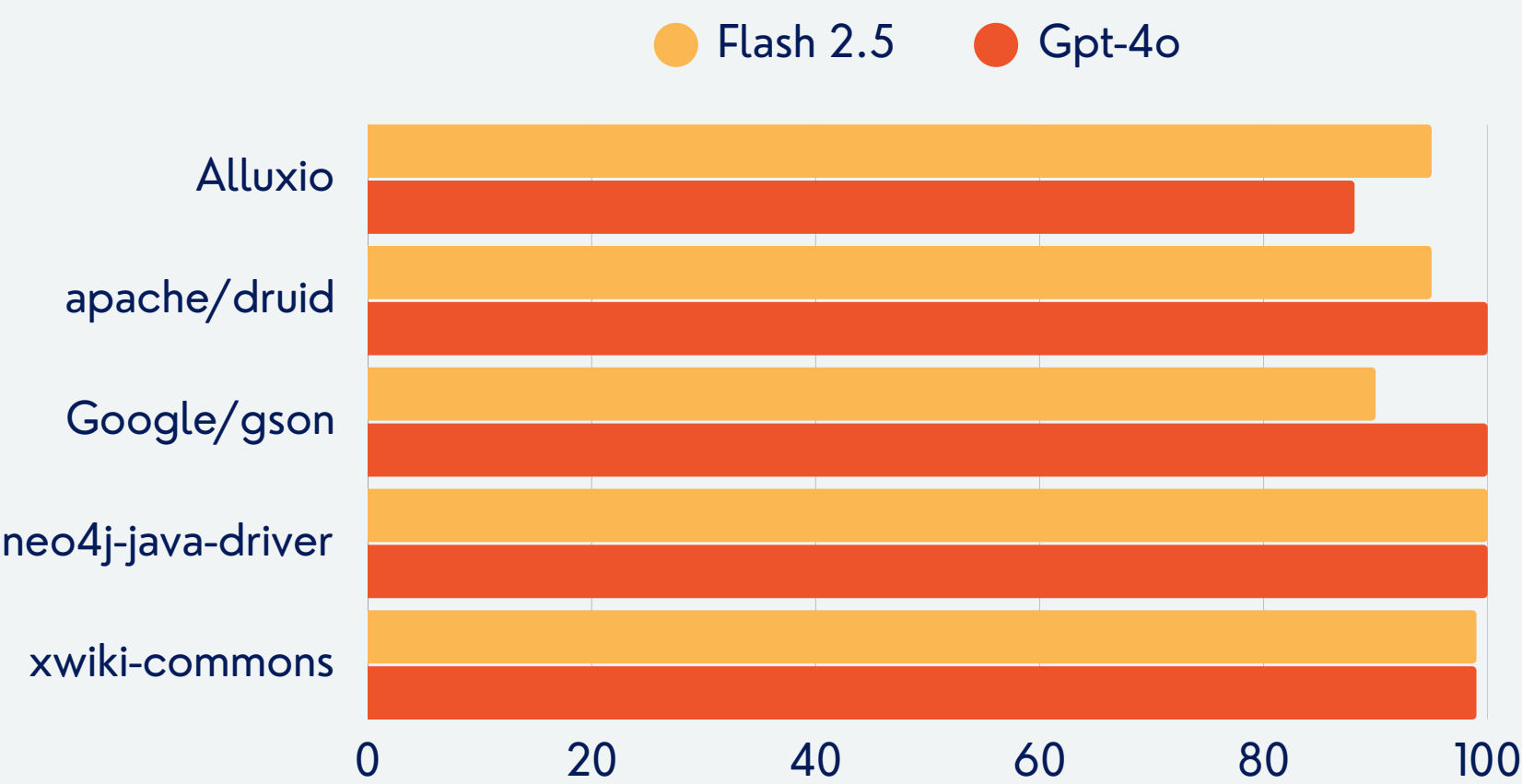


Insert Test v1 into project v2, it fails. Now we can start the repair via Agent. Test v2 will serve as a reference.

We have collected ~450 broken tests across 5 popular open-source projects

RESULTS

Succesfull repairs ratio per project



TEST SIMILARITY METRICS

Perfect Match: **70%**
Assertion Match: **85%**
CodeBleu: **~0.95**

WHAT'S NEXT?

- Evaluation vs Junie
- Test Repair vs Code repair oracle

